Felipe Branco de Paiva, MD

Computational Electrophysiology | Network Neuroscience | NeuroAI | Neuromodulation | Brain Dynamics

felipebrancopaiva@gmail.com | felipebpaiva.github.io | ORCID: 0000-0003-4225-6939

São Paulo, Brazil

Professional Summary

MD-trained electrophysiology and computational neuroscience researcher working on large-scale brain-network dynamics and computational modeling of neural activity. Experience spans rodent and non-human primate stereotactic surgery, analysis of neural time-series data, and interpretable machine learning and network-level modeling. Current work focuses on brain traveling waves, harmonic brain states, graph-based analysis, topological data analysis, and computational biomarkers for neurological and psychiatric diseases.

Research Focus

- **Computational electrophysiology:** Analysis of neural time-series data across MEG, EEG, SEEG, local field potentials, and spike recordings, using modality-appropriate preprocessing, source modeling, time-frequency analysis, and statistical modeling.
- **NeuroAI and interpretable modeling:** Application of topological data analysis, graph signal processing, machine learning, advanced signal processing methods, and computational modeling to multimodal brain data in health and disease.
- **Brain traveling waves:** Development of methods to characterize macroscopic brain traveling waves, focusing on graph-based decomposition approaches and advanced linear algebra.
- **Epilepsy and brain networks:** Graph-based analysis of brain-network organization in epilepsy, with emphasis on interpretable biomarkers, cognitive dysfunction, and network-level pathophysiology.
- **Neuromodulation and invasive neurophysiology:** Translational experience in DBS-related research, basal ganglia electrophysiology, surgical targeting, and preclinical models relevant to movement disorders and psychiatric disease.

Degrees & Education

Medical Doctor, MD 2009–2014
Faculdade de Ciências Médicas da Santa Casa de São Paulo *São Paulo, Brazil*

Medical education credentials complete and verified by ECFMG.

Research Appointments

Doctoral Researcher, Doctoral Programme Brain & Mind 2024–Present
University of Helsinki *Helsinki, Finland*
PI/Supervisor: Satu Palva, PhD; Matias Palva, PhD

- Analyze large-scale brain dynamics to characterize traveling waves in MEG data.
- Compare methods for characterizing macroscopic brain traveling waves, including graph-based decomposition approaches, linear phase fitting, two-dimensional fast Fourier transform, and computational modeling approaches.
- Improve MEG/EEG preprocessing pipelines, from artifact rejection and filtering to source reconstruction and inverse modeling, with emphasis on robust and reproducible Python implementation.
- Work within an interdisciplinary neuroscience environment integrating data analysis, computational modeling, cognitive neuroscience, and clinical research questions.

Postdoctoral Research Associate | American Epilepsy Society Postdoctoral Fellow 2023–2024
University of Wisconsin–Madison *Madison, WI, USA*
PI/Supervisor: Aaron Struck, MD; Moo K. Chung, PhD

- Awarded an American Epilepsy Society Postdoctoral Research Fellowship to support computational epilepsy research using high-density EEG and network-level modeling.
- Analyzed high-density EEG data in juvenile myoclonic epilepsy using topological data analysis, graph signal processing, machine learning, non-equilibrium dynamics, and biophysical modeling.
- Developed interpretable models of brain-network organization to investigate the pathophysiology of cognitive deficits in epilepsy.
- Led first-author work on thermodynamic rigidity of harmonic brain states in juvenile myoclonic epilepsy.
- Collaborated with clinicians and computational scientists to connect advanced analytical methods with clinically meaningful epilepsy phenotypes.

Research Specialist 2022–2023
University of Wisconsin–Madison *Madison, WI, USA*
PI/Supervisor: Melanie Boly, MD, PhD; Larissa Albantakis, PhD; Aaron Struck, MD

- Developed and maintained preprocessing and analysis pipelines for SEEG, EEG, MRI, and fMRI datasets.
- Performed data cleaning, quality control, organization, and reproducible analysis of multimodal clinical-neuroscience data.
- Supported epilepsy and systems-neuroscience projects requiring integration of electrophysiological and neuroimaging data.
- Worked across clinical and technical teams to transform heterogeneous datasets into research-ready analytical outputs.

Research Fellow 2017–2019
Cleveland Clinic, Lerner Research Institute *Cleveland, OH, USA*
PI/Supervisor: Andre Machado, MD, PhD; Kenneth Baker, PhD

- Performed advanced stereotactic procedures in non-human primates and contributed to preclinical neurophysiology studies of basal ganglia circuits.
- Analyzed spike activity and local field potential recordings to characterize electrophysiological signatures relevant to Parkinson's disease and DBS research.
- Led first-author work on the development and evaluation of a frameless stereotactic targeting system for subcortical nuclei in non-human primates.
- Presented DBS-related preclinical work at major neurosurgery and neuromodulation conferences.
- Gained translational experience connecting surgical targeting, invasive recordings, movement-disorder neurophysiology, and device-oriented research.

Visiting Researcher 2016–2017
University of Freiburg *Freiburg, Germany*
PI/Supervisor: Volker A. Coenen, MD; Mate Dobrossy, PhD

- Performed preclinical neurophysiology research involving surgical electrode implantation in rodents.
- Analyzed local field potential recordings from limbic and reward-related circuits in a rodent model of depression.
- Investigated electrophysiological biomarkers from the nucleus accumbens and related regions in the context of affective-disorder circuit dysfunction.
- Presented work at the World Society for Stereotactic and Functional Neurosurgery.

Research Funding & Awards

- **American Epilepsy Society Postdoctoral Research Fellowship**, 2023–2024. Amount: $50,000 USD for salary support. Duration: July 1, 2023–June 30, 2024. P.I./Applicant: Felipe Branco de Paiva, MD. Supervisor: Aaron Struck, MD.

Mentorship & Teaching

- **Undergraduate mentorship, University of Wisconsin–Madison:** Supervised and mentored undergraduate students Meixian Zhao, Meishu Zhao, and Steven Haworth in computational neuroscience research activities, including data organization, analysis workflows, programming concepts, and interpretation of electrophysiological datasets.
- **Programming and data-analysis teaching:** Designed and delivered informal programming tutorials for undergraduate students and medical doctors, focusing on practical scientific programming, reproducible data analysis, and computational approaches to biomedical research.

Publications & Preprints

1. **de Paiva FB**, Zhao M, Zhao M, et al. Thermodynamic rigidity of harmonic brain states relates to general mental ability in juvenile myoclonic epilepsy. *bioRxiv*. 2026. doi:10.64898/2026.04.06.715875.

2. David I, Sasai S, **de Paiva FB**, et al. Distance- and hierarchy-dependent functional dysconnectivity in schizophrenia and its association with cortical microstructure. *NeuroImage: Clinical*. 2026;49:103958. doi:10.1016/j.nicl.2026.103958.

3. Gorska-Klimowska U, Mat B, Denis C, et al. Multi-unit activity and high-gamma patterns dissociate in the seizure onset zone during focal to bilateral tonic-clonic seizures. *medRxiv*. 2025. doi:10.1101/2025.11.03.25339320.

4. Chung MK, Ramos CG, **de Paiva FB**, et al. Unified topological inference for brain networks in temporal lobe epilepsy using the Wasserstein distance. *NeuroImage*. 2023;284:120436. doi:10.1016/j.neuroimage.2023.120436.

5. **de Paiva FB**, Campbell BA, Frizon LA, et al. Feasibility and performance of a frameless stereotactic system for targeting subcortical nuclei in nonhuman primates. *Journal of Neurosurgery*. 2020;134(3):1064–1071. doi:10.3171/2019.12.JNS192946.

Selected Oral Presentations

1. **de Paiva FB**, Maldonado-Naranjo A, Campbell B, Machado AG, Baker KB. Novel Patterns of Deep Brain Stimulation (DBS): Preclinical Studies in the MPTP Non-Human Primate Model of Parkinson's Disease. *2019 AANS Annual Scientific Meeting*. April 13–17, 2019. San Diego, CA, USA.

2. **de Paiva FB**, Maldonado-Naranjo A, Campbell B, Machado AG, Baker KB. Efficacy and Mechanisms of Novel Patterns of Deep Brain Stimulation (DBS). *2018 Joint Meeting of Neuromodulation: The Science*. November 3–6, 2018. Cleveland, OH, USA.

Selected Poster Presentations

1. **de Paiva FB**, Chung MK, Struck AF. EEG Network Topology in Juvenile Myoclonic Epilepsy (JME). *2023 American Epilepsy Society Annual Meeting*. December 1–5, 2023. Orlando, FL, USA.

2. Monai E, **de Paiva FB**, Riedner B, Sevak B, Tononi G, Boly M, Baird B. Inception: Phase-Amplitude Coupling Predicts Incorporation of External Stimuli into Dreams. *Annual Meeting of the Society for Neuroscience*. November 11–16, 2022. San Diego, CA, USA.

3. **de Paiva FB**, Mottaghi S, Dobrossy MD, Coenen VA. Dynamics of Local Field Potential Activity Recorded in the Ventral Tegmental Area and the Nucleus Accumbens in a Rodent Model of Depression. *17th Quadrennial Meeting of the World Society for Stereotactic and Functional Neurosurgery*. June 26–29, 2017. Berlin, Germany.

Technical & Methodological Expertise

Programming & environments	Python, MATLAB, Julia, Linux/Bash, high-performance computing environments
Machine learning & modeling	PyTorch/CUDA, Scikit-learn, interpretable modeling, graph-based methods, biophysical modeling
Electrophysiology analysis	MEG, EEG, high-density EEG, SEEG, local field potentials, spike activity, source modeling, wavelet-based time-frequency analysis
Network & graph methods	Graph signal processing, network analysis, harmonic decomposition, generalized eigendecomposition, graph-based modeling of brain dynamics
Topological methods	Topological data analysis, Wasserstein-distance-based network inference, topology-informed analysis of brain networks
Neuroimaging pipelines	MRI/fMRI preprocessing and analysis, SPM, fMRIPrep, multimodal EEG/MRI/fMRI integration
Neuroscience toolboxes	MNE, EEGLAB, FieldTrip, SPM, fMRIPrep, HNN
Scientific communication	Manuscript writing, conference presentations, grant/fellowship applications, interdisciplinary scientific communication

Languages

Portuguese: native | English: fluent | German: intermediate | Spanish: intermediate

References

Available upon request.